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MAT 303 — Past Questions & Materials

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Course: MAT 303

Institution: Federal University of Technology, Akure

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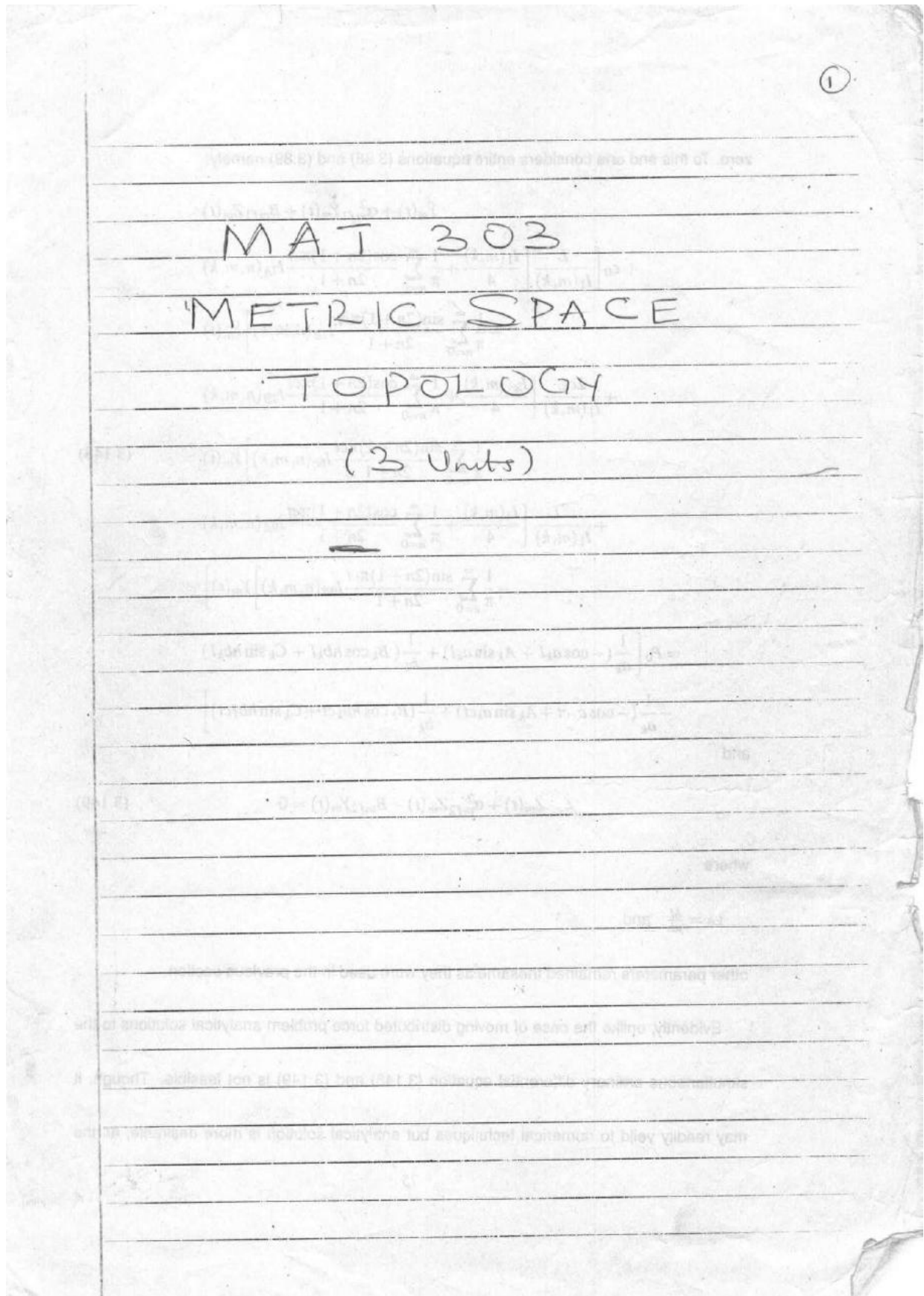
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Paper 1 of 1 — Cyssa hubMAT 303 - METRIC SPACE TOPOLOGY N 001 (1).pdf

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MAT 303 (3 units)

①

METRIC SPACE TOPOLOGY

Course Outline

Sets, metrics, and examples

Open spheres (or balls)

Open sets and neighbourhoods.

Closed sets

Interior, exterior, frontier, limit points.
and closure of a set.

Dense subsets and Separable space

Convergence in metric spaces, homeomorphism.

Continuity and Compactness, Connectedness.

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SETS

DEFINITION: Sets, Elements

1. A set is any well-defined list or collection of related objects.
2. An object in a set is called an element or a member of the set.

Notation and Representation

Use capital letters $P, Q, R, S, \dots$ to represent sets and lower case letters $p, q, r, s, \dots$ to represent elements of a set.

$x \in P$ denotes x is a member or an element of set P .

$y \notin P$ denotes y is not a member or an element of set P .

Description of Sets

There are essentially different ways of specifying a particular set.

(i) Using a diagram: A set can be described by drawing the objects in the set in a diagram.

(ii) By Word description: A set can be described in words. For example, a set containing numbers 1, 2, 3, 4, 5 can be described as a set of natural or counting numbers from 1 to 5.

(3)

(iii) By listing the elements: A set can be described by listing the elements inside a pair of curly brackets or braces. For example, a set consisting numbers 1, 2, 3, 4, 5 is described as $\{1, 2, 3, 4, 5\}$.

(iv) Sample element description: Another way of describing the set is by stating those properties which characterize the elements in the set. For example;

$$A = \{x : x \text{ is an integer, } x > 0\}$$

which reads A is a set of x such that x is an integer and x is greater than zero. The set A above can be re-written as;

$$A = \{1, 2, 3, \dots\}$$

This denotes A is a set whose elements are positive integers.

Note that intervals on the real line, defined below, appear very often in Mathematics. Here, a and b are real numbers with $a < b$.

Open interval from a to $b = (a, b)$

$$= \{x : a < x < b\}$$

Closed interval from a to $b = [a, b] = \{x : a \leq x \leq b\}$

$$= \{x : a \leq x \leq b\}$$

Open-closed interval a to $b = (a, b]$

$$= \{x : a < x \leq b\}$$

(4)

Closed - open interval from a to b
 $= [a, b) = \{x : a \leq x < b\}$.

Types of Sets

(i) Null or Empty Set

The set containing no element is called the empty set. It is denoted by $\{\}$ or ϕ .

(ii) Singleton Set

A set containing only one element

(iii) Finite and Infinite Sets

If the number of elements in a set is finite (i.e. positive integer)

the set is called a finite set. The number of elements in a finite set A is denoted by $n(A)$.

~~A set containing infinite number of~~ e.g. $A = \{a, b, c, d, e, f\}$

Here, $n(A) = 6$

A set consisting an infinite number of element is called an infinite set.

For example the following sets are infinite.

(a) $A = \{x : x \text{ is a natural number}\}$

$\Rightarrow A = \{1, 2, 3, 4, \dots\}$

(b) $B = \{x : x \text{ is an integer}\}$

$\Rightarrow B = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

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(VII) SUBSETS, SUPER SETS.

A set A is a subset of a set B or written $A \subseteq B$ iff each element in A also belongs to B i.e. $x \in A$ implies $x \in B$. In essence, we say A is contained in B . Equivalently, a set B is a superset of A written $B \supseteq A$ i.e. Set B contains every element of set A .

Example:

Consider the sets

$$A = \{1, 3, 5, 7, 11, \dots\}, \quad B = \{5, 10, 15, 20, \dots\}$$

$$C = \{x : x \text{ is prime, } x > 2\} = \{3, 5, 7, 11, \dots\}$$

Here, $C \subseteq A$ and $B \not\subseteq A$.

Theorem:

Let A , B and C be any sets. Then,

- (i) $A \subseteq A$
- (ii) If $A \subseteq B$ and $B \subseteq A$, then $A = B$
- (iii) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

NOTE:

- (i) Every set is a subset of itself.
- (ii) Empty set is a subset of every set.

If A is a subset of B , but there is at least one element/member of B , which is not in A , then A is called proper subset of B written $A \subset B$.

⑦

(vii) UNIVERSAL SET

The set containing all elements under discussion is called the universal set. It is denoted by U or E .

Examples:

(a) In plane geometry, the universal set consists of all the points in the plane.

(b) Let $A = \{x : x^2 = 4, x \text{ is odd}\}$.
Then, A is empty i.e. $A = \phi$.

(c) Let $B = \{\phi\}$. Then, $B \neq \phi$ (i.e. B contains one element).

CLASSES, COLLECTIONS, FAMILIES AND SPACES

The members of a set are sets themselves. We use the words "class", "collection" and "family" synonymously with set.

We use CLASS for a set of sets and COLLECTION OR FAMILY for a set of classes.

Example:

$A = \{\{2,3\}, \{2\}, \{5,6\}\}$ is a class and its members are the set $\{2,3\}$, $\{2\}$ and $\{5,6\}$.

The words "Subclass", "sub collection" and "Subfamily" have meanings analogous to subset. While the word "Space" means a non-empty set which possesses some types of mathematical structure, e.g. vector space, metric space or topological space.

⑧

POWER OF SET A (denoted by $P(A)$)

This is the class of all subsets of set A.
(i.e. the set of all subsets of set A).

Example:

If $A = \{a, b, c\}$, the subsets of A are $\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \phi$, and $\{a, b, c\}$.

Then, $P(A) = \{ \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \phi, \{a, b, c\} \}$

In general, if A is finite set, say A has n elements, then $P(A)$ will have 2^n elements.

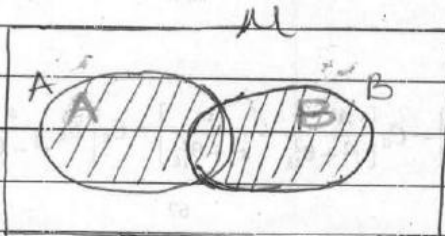
SET OPERATIONS

(i) Union of Sets

The union of two sets A and B denoted by $A \cup B$, is the set of all elements which belong to A or B.

i.e. $A \cup B = \{x : x \in A \text{ or } x \in B\}$

Here, "or" is used in the sense of "and/or".



Venn diagram of $A \cup B$ (shaded)

(9)

Theorem :

For any Set A

(a) $A \cup A = A$

(b) $A \cup \phi = A$

(c) $A \cup U = U$

(d) If $A \subseteq B$, then $A \cup B = B$

Example :

If $A = \{1, 2, 3, 5, 6\}$ and $B = \{2, 5, 7, 8, 9\}$ Then $A \cup B = \{1, 2, 3, 5, 6, 7, 8, 9\}$

(ii) Intersection of Sets (A and B)

The intersection of two sets A and B is the set of elements which belong common to A and B. It is denoted by $A \cap B$.

i.e. $A \cap B = \{x : x \in A \text{ and } x \in B\}$

If $A \cap B = \phi$, then A and B are said to be disjoint or non-intersecting.

A class A of sets is called a disjoint class of sets if each pair of distinct sets in A is disjoint.

Theorem : For any set A

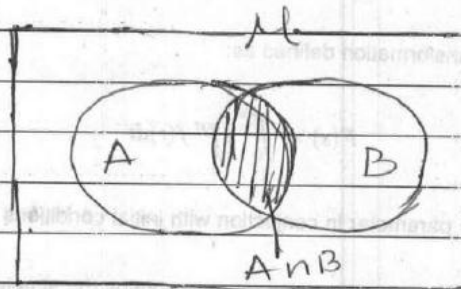
(a) $A \cap A = A$

(b) $A \cap \phi = \phi$

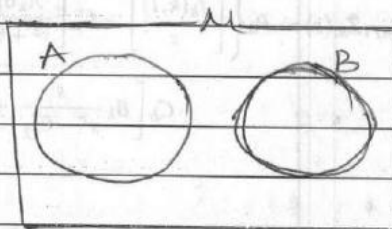
(c) $A \cap U = A$

(d) If $A \subseteq B$, then $A \cap B = A$

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The Venn diagram of $A \cap B$ (shaded)



Sets A and B are disjoint. i.e. $A \cap B = \phi$

Example:

Given three sets

$$A = \{-1, 5, 7, 8\}, B = \{0, 2, 5, 6, 8\}$$

$$\text{and } C = \{-3, -5, 1, 7\}$$

Then, $A \cap B =$

$$\text{Then, } A \cap B = \{5, 8\}$$

$$A \cap C = \{1, 7\} \text{ and}$$

$$B \cap C = \phi$$

(iii) Set difference (ie Relative Complement)

The set difference A and B denoted by $A \sim B$ or $A \setminus B$ is defined as

$$A \setminus B = \{x: x \in A, x \notin B\}$$

go to pg 331.

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(iv) Equal Sets

Two sets A and B are equal (denoted $A=B$) if they contain identically the same elements, (not necessarily in the same order).
Thus, $A=B$ means $A \subseteq B$ and $B \subseteq A$
i.e. All members of A are contained in B and all members of B are contained in A .

Example:

Given three sets, $A = \{x : x^2 - 3x + 2 = 0\}$,
 $B = \{2, 1\}$ and $C = \{1, 2, 2, 1\}$.

$$\begin{aligned}\text{Thus, } A &= \{x : x^2 - 3x + 2 = 0\} \\ &= \{(x-1)(x-2) = 0\} \\ \Rightarrow A &= \{1, 2\}\end{aligned}$$

Then, $A = B = C$.

Note that a ^{Set} A remains the same if its elements are repeated or rearranged.

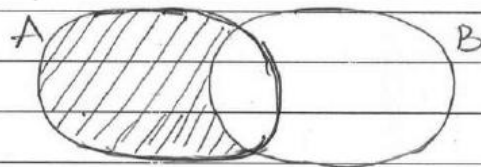
(v) Equivalent Sets

Two sets A and B are equivalent, if we can assign a different member of B to every member of A so that every member of B gets assigned to one and only one member of A .

$A \approx B$ denote A is equivalent to B .

If A and B are finite sets and $n(A) = n(B)$ then A and B are equivalent.

(11)



$A \setminus B$ is shaded

Example:

If $A = \{0, 3, 4, 8, 9, 13\}$ and $B = \{1, 2, 3, 4, 8, 9\}$

then, $A \setminus B = \{0, 13\}$

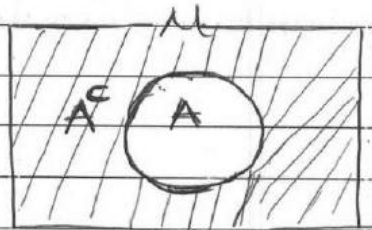
$B \setminus A = \{1, 2\}$ and

$(A \setminus B) \cap (B \setminus A) = \emptyset$

(IV) Complement of a set (Absolute Complement of a set)

If A is any set in a universal set U , then the complement of a set A denoted by A^c or A' is defined as a set

$$A^c = \{x : x \in U, x \notin A\}$$



A^c is shaded

Example

If $A = \{a, b, d, g\}$ and $B = \{b, c, g, f, h\}$ are subsets of universal set $U = \{a, b, c, d, e, f, g, h, i, j\}$

then, $A^c = \{c, e, f, h, i, j\}$

$B^c = \{a, d, e, i, j\}$

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Symmetric difference of two sets

Let A and B be sets. The Symmetric difference of A and B , written $A \Delta B$, is defined as a set

$$A \Delta B = (A \setminus B) \cup (B \setminus A)$$

Example

If $A = \{1, 2, 5, 4, 7, 9\}$ and

$$B = \{2, 3, 4, 6, 8\}$$

Then, $A \setminus B = \{1, 5, 7, 9\}$

$$B \setminus A = \{3, 6, 8\}$$

Hence, $A \Delta B = A \setminus B \cup B \setminus A$

$$= \{1, 3, 5, 6, 7, 8, 9\}$$

Finite Sets and Venn diagrams

- For any two finite sets A and B

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

- For any three (3) finite sets A , B and C

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Question:

(1) In a college of 200 students, 140 students like tea, 120 students like coffee and 80 students like both tea and coffee.

(i) How many students like only tea?

(i) tea only? (ii) coffee only? (iii) neither tea nor coffee?

(2) Let $A = \{1, 2, 3, 5, 7, 9, 18, 20\}$ and $B = \{3, 4, 5, 12, 13, 15, 17\}$ be subsets of universal set $U = \{1, 2, 3, 4, \dots, 20\}$

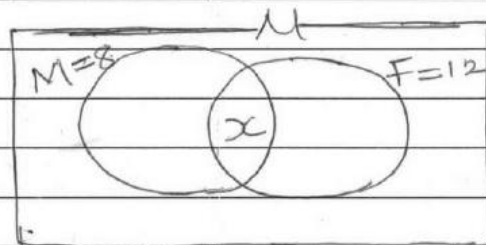
Verify $(A \cup B)^c = A^c \cap B^c$.

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Question

In a class, 8 boys study metal work and 12 boys study French. Altogether 15 boys study French or metal work or both.

- (a) Draw a Venn diagram to illustrate the information.
 (b) Find out how many boys study both subjects.

Solution

$$n(M) = 8$$

$$n(F) = 12$$

$$n(M \cup F) = 15$$

$$n(M \cap F) = x$$

Recall, $n(M \cup F) = n(M) + n(F) - n(M \cap F)$

$$\text{i.e. } 15 = 8 + 12 - x$$

$$x = 20 - 15 = 5$$

Que

Let P and Q be two finite sets such that $n(P \cup Q) = 54$, $n(P \cap Q) = 8$ and $n(Q) = 27$. Find $n(P)$.

Solution

Given, $n(P \cup Q) = 54$, $n(P \cap Q) = 8$, $n(Q) = 27$, $n(P) = ?$

$\Rightarrow$ Recall, $n(P \cup Q) = n(P) + n(Q) - n(P \cap Q)$

$$\text{i.e. } 54 = n(P) + 27 - 8$$

$$n(P) = 54 - 19 = 35$$

LAWS OF THE ALGEBRA OF SETS

① Idempotent laws

$$(1a) \quad A \cup A = A$$

$$(1b) \quad A \cap A = A$$

② Commutative laws

$$(2a) \quad A \cup B = B \cup A$$

$$(2b) \quad A \cap B = B \cap A$$

③ Associative Laws

$$(3a) \quad (A \cup B) \cup C = A \cup (B \cup C)$$

$$(3b) \quad (A \cap B) \cap C = A \cap (B \cap C)$$

④ Distributive Laws

$$(4a) \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(4b) \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

⑤ Identity Laws

$$(5a) \quad A \cup \phi = A \quad (\phi \text{ is an empty set})$$

$$(5b) \quad A \cup U = U \quad (U \text{ is a universal set})$$

$$(6a) \quad A \cap \phi = \phi$$

$$(6b) \quad A \cap U = A$$

⑦ Complement Laws

$$(7a) \quad A \cup A^c = U \quad (7b) \quad A \cap A^c = \phi$$

$$(8a) \quad (A^c)^c = A \quad (8b) \quad U^c = \phi, \phi^c = U$$

⑨ De Morgan's Laws

$$(9a) \quad (A \cup B)^c = A^c \cap B^c$$

$$(9b) \quad (A \cap B)^c = A^c \cup B^c$$

(12)

Cartesian Product of sets (Product Set)

Let A and B be two sets. The cartesian product or simply product sets of A and B , written $A \times B$, consists of all ordered pairs (a, b) where $a \in A$ and $b \in B$,

$$i.e. A \times B = \{(a, b) : a \in A, b \in B\}$$

The product of a set with itself, say $A \times A$, will be denoted by A^2 .

Example

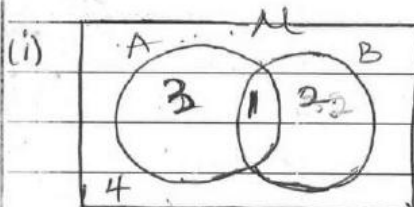
Let $A = \{2, 3, 4\}$ and $B = \{p, q\}$

Then, $A \times B = \{(2, p), (2, q), (3, p), (3, q), (4, p), (4, q)\}$

$B \times A = \{(p, 2), (p, 3), (p, 4), (q, 2), (q, 3), (q, 4)\}$

Question!

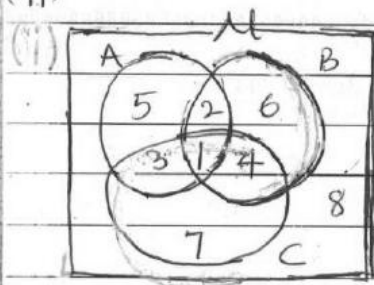
Name the regions 1 to 10 in the Venn diagrams given below.



Solution

Region	1	2	3	4
Name	$A \cap B$	$A^c \cap B$	$A \cap B^c$	$A^c \cap B^c$

(ii)



Solution

Region	Name	Description
1	$A \cap B \cap C$	Intersection of A, B & C
2	$A \cap B \cap C^c$	A & B only
3	$A \cap B^c \cap C$	A & C only
4	$A^c \cap B \cap C$	B & C only
5	$A \cap B^c \cap C^c$	A only
6	$A^c \cap B \cap C^c$	B only
7	$A^c \cap B^c \cap C$	C only
8	$(A \cup B \cup C)^c$	Complement of $(A \cup B \cup C)$

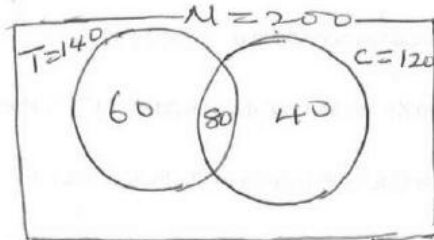
NOTE

$$(A \cup B \cup C)^c = A^c \cap B^c \cap C^c$$

(14)

(15)

$$M = 200 \quad n(T) = 140, \quad n(C) = 120, \quad n(T \cap C) = 80$$



(i) Those who like only Tea.

$$n(T \text{ only}) = 140 - 80 = 60 \text{ students}$$

(ii) Those who like only Coffee.

$$n(C \text{ only}) = 120 - 80 = 40 \text{ students}$$

(iii) Those who like neither tea nor coffee.

$$\begin{aligned} n(T \cup C)^c &= M - [n(T \text{ only}) + n(C \text{ only}) + n(T \cap C)] \\ &= 200 - (60 + 40 + 80) \\ &= 200 - 180 = 20 \text{ students} \end{aligned}$$

(5b)

Given, $M = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$

$A = \{1, 2, 3, 5, 7, 9, 18, 20\}$

$B = \{3, 4, 5, 12, 13, 15, 17\}$

$$A \cup B = \{1, 2, 3, 4, 5, 7, 9, 12, 13, 15, 17, 18, 20\}$$

$$(A \cup B)^c = \{6, 8, 10, 11, 14, 16, 19\}$$

$$A^c = \{4, 6, 8, 10, 11, 12, 13, 14, 15, 16, 17, 19\}$$

$$B^c = \{1, 2, 6, 7, 8, 9, 10, 11, 14, 16, 18, 19, 20\}$$

$$A^c \cap B^c = \{6, 8, 10, 11, 14, 16, 19\}$$

$$\text{Hence, } (A \cup B)^c = A^c \cap B^c \quad \checkmark$$

MAT 101

(17)

Exercises on Sets

Exercise 1

(1) Which of the following is / are subset(s) of $\{a, e, i, o, u\}$

- (i) $\{a, e, i, n\}$ (ii) $\{a, i, u\}$ (iii) $\{m, a, e, u\}$
 (iv) $\{a, e, i, t\}$ (v) $\{u, e, i, o, a\}$ (vi) ϕ

(2) Write the set $\{x: -3 < x \leq 5, x \text{ is an integer}\}$ by listing the elements

(3) Write down all the subsets of
 (a) $\{3, 7\}$ (b) $\{a, e, i\}$

(4) List the member(s) of the following sets.

- (a) $\{x: x+3=9\}$ (b) $\{x | x^2=1\}$
 (c) $\{x: x^2+1=0, x \text{ is a real number}\}$
 (d) $\{x | x^2=16 \text{ and } 3x=9\}$

(5) If $U = \{2, 4, 6, 8, \dots, 20\}$ is the universal set, write down by listing the elements, the sets:

- (a) $\{\text{Perfect square}\}$
 (b) $\{\text{Multiples of } 8\}$
 (c) $\{\text{numbers greater than } 10\}$

(6) If $A = \{a, e, u\}$ and $B = \{6, 7, 8, 9\}$
 Find:

- (a) $A \times B$ (b) $B \times A$
 (c) $n(A \times B)$ (d) $n(B \times A)$

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Exercise 2:

- ① A and B are subsets of a universal set U , and are not disjoint.

Represent them by means of a Venn diagram. Shade the regions which represent the set:

- (i) $A \cap B^c$ (ii) $B \cap A^c$ (iii) $(A \cap B^c) \cup (A^c \cap B)$

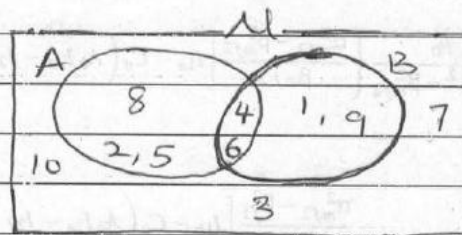
- ② Show, using the Venn diagrams, that

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(ii) $(A \cup B)^c = A^c \cap B^c$

(iii) $(A \cap B)^c = A^c \cup B^c$

- ③ Consider the Venn diagram given below



List the element(s) of the set:

- (i) A^c (ii) B^c (iii) $A^c \cup B^c$
 (iv) $A^c \cap B^c$ (v) $(A \cup B)^c$ (vi) $(A \cap B)^c$

- ④ If $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$, find:

- (i) $A \setminus B$ (ii) $C \setminus A$ (iii) $B \setminus C$
 (iv) $(A \setminus B) \cap (B \setminus A)$ (v) $(A \setminus B) \cup (B \setminus C)$
 (vi) $(A \setminus B) \cap (C \setminus A) \cap (B \setminus C)$
 (vii) $(A \setminus B) \cup (C \setminus A) \cup (B \setminus C)$

(19)

Exercise 3

① Each of 39 of my friends has either a dog, a cat or a rabbit. Each of 24 of them has a dog, each of 17 has a cat and each of 16 has a rabbit. The number having both a dog and a cat is one (1) more than the number having both a cat and a rabbit. There are nine (9) who have both a dog and a rabbit while two (2) of them have all the three.

How many of my friends have both a rabbit and a cat?

(ANS: $n(R \cap C) = 5$)

② In a group of 20 students, 9 like bread, but not beans while 14 like bread. How many like beans but not bread?

③ A company employs 100 people, 65 of whom are men. 60 people including all women are paid weekly. How many of the men are paid weekly?

④ Of a group of 200 persons, 100 are interested in Music, 70 are interested in Photography and 40 like Skiing. Also, 40 are interested in both Music and Photography, 30 in both Music and Skiing. 20 in both Photography and Skiing, while 20 are interested in Photography but not in Music or Skiing. How many persons are interested in all the three?